MIL-E-45332C(AT)
30 June 1975
SUPERSEDING
MIL-E-45332B(AT)
26 May 1969

MILITARY SPECIFICATION

ENGINE GASOLINE: 4-CYLINDER, IN-LINE
LIQUID-COOLED, 65 HP, M151-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the type of 4-cylinder, liquid-cooled, 4-stroke cycle, naturally aspirated internal combustion, gasoline engine for use in military vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-A-548	-	Antifreeze, Ethylene Glycol, Inhibited.
O-I-490	-	Inhibitor, Corrosion, Liquid Cooling System.
VV-G-001690	-	Gasoline, Automotive.

Military

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine (Heavy Duty).
MIL-L-10295	-	Lubrication Oil, Internal Combustion Engine, Sub-Zero.
MIL-A-11755	-	Antifreeze; Arctic-Type.

FSC-2805

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MIL-A-13488 - Air Cleaner Engine: Heavy Duty, Oil-Bath Type (for Internal Combustion Engines).
MIL-E-13856 - Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-46015 - Gasoline, Automotive, Combat, Referee Grade.
MIL-C-52412 - Cleaning Compound, High Pressure Cleaner, Steam.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

DRAWINGS
Army
11613729 - Engine.
11660437 - Power Plant Assembly, M151A2.
8712420 - Army Winterization Kit.

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS
SQAP-INDEX
11613729 - Supplemental Quality Assurance Provisions.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF COMMERCE

(Publication No. 235 - Weather Bureau Psychrometric Tables.)

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C., 20402.)

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3. REQUIREMENTS

3.1 Initial production engine. The initial engine produced shall be submitted to the Government and shall be subjected to examination and test to determine conformance to the initial production engine provisions of this specification (see 4.2 and 6.3). The initial production engine shall be fully representative of the engines to be supplied from production facilities and tooling.

3.2 500 hour endurance test (Initial production engine). The initial production engine using fuel conforming to VV-G-001690, shall be operated for a period of not more than 575 hours. At the conclusion of a 500-hour minimum endurance test (Table I) the engine shall meet the operational and performance requirements specified herein. During this run the engine shall require only occasional, normal military maintenance such as replacement of ignition contact points, spark plugs, oil filters, lubricating oil and necessary adjustments. At the conclusion of this period, the engine shall demonstrate the performance requirements specified herein with the exdeption that the engine shall develop not less than 90 percent of the originally specified horsepower and torque.

3.2.1 500-hour endurance test set-up. The initial production engine, as received from the contractor at the location designated by the procuring activity in the purchase contract shall be prepared for the endur-ance test as follows:

(a) The flywheel housing and transmission torus cover shall be removed. A special wing plate incorporating rubber shock mountings adaptable to the test cell engine mounting pedestals, shall be attached to the rear face of the crankcase in place of the flywheel housing.
(b) Special rubber shear mountings, adaptable to the test cell engine mounting pedestals, shall be installed on the front engine mount bolt pads located on either side of the crankcase.
(c) Thermocouples, plumbing and other equipment necessary for monitoring data during the test, shall be installed in such a manner as to not impair the normal functioning of the engine.
(d) In the absence of the engine cooling fan and radiator, a suitable "house" coolant supply system shall be substituted that is capable of flow rate as well as temperature regulation to assure a constanct supply of coolant to the engine at the specified temperatures.
(e) A regulated cold water spray system shall be installed as a means of regulating engine oil temperature during the test.
(f) Heat shields shall be installed between the exhaust manifolds and the rocker arm covers on the engine to prevent radiant heat damage to the rocker arm cover gaskets.
(g) An auxiliary cooling air supply shall be directed toward the front end of the engine to aid in carrying away excessive heat that would normally be handled by the engine cooling fan.

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(h) The engine, after preparation for the test, shall be inspected and approved by a representative of the contractor.

3.2.2 Test temperatures. The final 100 hours of the 500-hour endurance test shall be run at ambient test cell temperatures of 115° ± 5°F under simulated maximum solar radiation exposure conditions and relative humidity of five percent. During the period the oil temperature, measured in the engine oil sump, shall not exceed 300°F. The coolant temperature, measured at the engine outlet, shall be maintained at 200° ± 5°F. The first 400 hours and all power check runs shall be run at ambient test cell temperatures sufficient to control the carburetor air inlet temperature to not less than 60°F nor more than 100°F. The coolant outlet temperature shall be maintained at 180° ± 5°F during these portions of the test.

3.2.3 Test support. The contractor shall expeditiously furnish spare parts to the Government as required to support the 500-hour endurance test.

3.2.4 Servicing. Prior to each power check test specified in the 500-hour endurance run schedule, the lubricating oil and oil filter shall be changed, spark plugs and ignition points checked and changed as necessary and the ignition points and timing adjusted to applicable specifications.

3.2.5 Adjustments and/or checks. With reference to table I, engine adjustments and/or checks as noted below will be accomplished using start or run #2 as zero time.

(a) Twenty-five (25) hour checks:

(1) All engine valves to be checked for lash and reset to specification, if necessary.
(2) Check intake manifold vacuum. If vacuum gauge indicates less than 17.0 in Hg. at 650 engine RPM, check individual cylinders to locate problem area. Analyze by checking spark plugs and head gasket for leakage, head bolts for lesser torque and also valves for proper clearance. Replace faulty spark plugs only. If valve clearance is problem area, replace defective valve (valves) and regrind replaced valve seat (seats) to drawing requirements.

(b) Fifty (50) hour checks:

(1) Check spark plugs for fouling and gap. Clean and reset to specification, if necessary.
(2) Check distributor point gap. Clean and reset to specification, if necessary. Check the ignition timing.

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(c) One Hundred (100) hour check:

(1) Check spark advance. Reset to specification, if necessary.

(d) Two Hundred Fifty (250) hour check:

(1) Replace distributor points and spark plugs. Check the ignition timing.

3.2.6 Cooling conditions. Circulation of the coolant through the engine shall be accomplished with the engine coolant pump. The coolant thermostat shall be removed or blocked in the full open position. Refer to Para. 3.3.5 for coolant requirements.

3.2.7 Oil. Referee grade oil shall be used for the entire 500-hour endurance test (see 3.3.4).

3.2.8 Fuel. Fuel conforming to VV-G-001690 shall be used for the entire 500-hour endurance test (see 3.2).

3.2.9 Test hours. Except where shorter periods are specified in the 500-hour endurance run schedule, warm-up, starting and stopping times shall be recorded; however, warm-up time and running time of less than 30 minutes duration shall not be counted toward fulfillment of the test run hour requirements.

3.2.10 Required data. During the 500-hour endurance test, the following data shall be recorded at the end of each hour of operation, at the completion of each period, and just prior to stopping the engine (normal shut down only: emergency stops excluded). After stopping and restarting, the engine shall be allowed to warm-up for a sufficient length of time, at the scheduled speed for that period, to stabilize speed, torque and temperatures before data is recorded:

(a) Engine speed (rpm).
(b) Observed brake horsepower (observed bhp).
(c) Observed torque (observed lbs ft).
(d) Inlet manifold vacuum (in Hg).
(e) Exhaust manifold pressure (in Hg).
(f) Fuel pressure (psi at carburetor).
(g) Lubricating oil pressure (psi at rear gallery opening).
(h) Crankcase pressure (in Hg).
(i) Lubricating oil temperature (^{o}F at sump drain plug).
(j) Fuel temperature (^{o}F in carb float bowl).
(k) Carburetor air inlet temperature (^{o}F).
(l) Coolant temperature (^{o}F inlet and outlet).
(m) Exhaust temperature (^{o}F at exhaust manifold outlet).

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(n) Test cell ambient temperature (°F).
(o) Blow-by (cfm-crankcase).
(p) Oil consumption (lbs/hr - readings at 4-hour intervals).
(q) Fuel consumption (lbs/hr).
(r) Specific fuel consumption (lbs/observed bhp hr).
(s) Specific oil consumption (lbs/observed bhp hr).
(t) Barometric pressure (in Hg - readings at 4-hour intervals).
(u) Relative humidity (psychrometer, psychrometric table - readings at 4-hour intervals).

3.2.11 Disposition of engine. After completion of the 500-hour endurance tests, the engine shall be completely disassembled and all components subjected to a detail inspection. A detailed report of this inspection shall be accomplished either at the contractor's plant or other location as designated in the contract or purchase order. If the contract specified the contractor's plant for this activity, the engine will be shipped to his location for disassembly and inspection in the presence of those Government representatives so designated by the contracting officer. After completion of the inspection to the satisfaction of the Government, the engine shall be either disposed of at the contractor's plant or reassembled and shipped as a used engine to location designated by the contracting officer.

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Table I. 500 hour endurance run schedule

Run	Time (hours)	Engine Speed (rpm)	Load (Observed) hp	Coolant Temp. °F
1	Run-in schedule - see table I			
2	2	650	0	180
3	5	1800	30	180
4	1	650	0	180
5	5	3000	49	180
6	1	650	0	180
7	5	3500	52	180
8	1	650	0	180
9 *	5	4000	53	180
10	2	650	0	180
11	5	1800	30	180
12	1	650	0	180
13	5	3000	49	180
14	1	650	0	180
15	5	3500	52	180
16	1	650	0	180
17 *	5	4000	53	180
18	2	650	0	180
19	5	1800	30	180
20	1	650	0	180
21	5	3000	49	180
22	1	650	0	180
23	5	3500	52	180
24	1	650	0	180
25 *	5	4000	53	180
26	2	650	0	180
27	5	1800	30	180
28	1	650	0	180
29	5	3000	49	180
30	1	650	0	180
31	5	3000	52	180
32	1	650	0	180
33 *	5	4000	53	180
34	Power check run - see paragraph 3.5.1			
35	2	650	0	140
36	5	1800	30	140
37	1	650	0	140
38	5	3000	49	140
39	1	650	0	140
40	5	3500	52	140
41	1	650	0	140
42 *	5	4000	53	140
43	2	650	0	140
44	5	1800	30	140

* 25 Hour Intervals

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Table I. 500 hour endurance run schedule (cont'd)

Run	Time (hours)	Engine Speed (rpm)	Load (Observed) hp	Coolant Temp. °F
45	1	650	0	140
46	5	3000	49	140
47	1	650	0	140
48	5	3500	52	140
49	1	650	0	140
50 *	5	4000	53	140
51	2	650	0	180
52	5	1800	30	180
53	1	650	0	180
54	5	3000	49	180
55	1	650	0	180
56	5	3500	52	180
57	1	650	0	180
58 *	5	4000	53	180
59	2	650	0	180
60	5	1800	30	180
61	1	650	0	180
62	5	3000	40	180
63	1	650	0	180
64	5	3500	52	180
65	1	650	0	180
66 *	5	4000	53	180
67 Power check run - see paragraph 3.5.1				
68	2	1800	22	180
69	1	650	0	180
70	2	1800	WOT	180
71	2	1800	22	180
72	1	650	0	180
73	2	1800	WOT	180
74	2	1800	22	180
75	1	650	0	180
76	2	1800	WOT	180
77	2	1800	22	180
78	1	650	0	180
79	2	1800	WOT	180
80	2	1800	22	180
81	1	650	0	180
82 *	2	1800	WOT	180
83	2	1800	22	180
84	1	650	0	180
85	2	1800	WOT	180
86	2	1800	22	180
87	1	650	0	180

* 25 Hour Intervals

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Table I. 500 hour endurance schedule (cont'd)

Run	Time (hours)	Engine Speed (rpm)	Load (Observed) hp	Coolant Temp. °F
88	2	1800	WOT	180
89	2	1800	22	180
90	1	650	0	180
91	2	1800	WOT	180
92	2	1800	22	180
93	1	650	0	180
94	2	1800	WOT	180
95	2	1800	22	180
96	1	650	0	180
97 *	2	1800	WOT	180
98	4	3000	22	180
99	1	650	0	180
100	4	3000	WOT	180
101	1	650	0	180
102	4	3000	33	180
103	1	650	0	180
104	4	3000	WOT	180
105	1	650	0	180
106	4	3000	33	180
107 *	1	650	0	180
108	4	3000	WOT	180
109	1	650	0	180
110	4	3000	33	180
111	1	650	0	180
112	4	3000	WOT	180
113	1	650	0	180
114	4	3000	33	180
115	1	650	0	180
116	4	3000	WOT	180
117	1	650	0	180
118 Power check run - see paragraph 3.5.1				
119	4	3500	WOT	180
120	1	650	0	180
121	4	3500	52	180
122	1	650	0	180
123	4	3500	WOT	180
124	1	650	0	180
125	4	3500	52	180
126	1	650	0	180
127	4	3500	WOT	180
128 *	1	650	0	180
129	4	3500	52	180

* 25 Hour Intervals

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Table I. 500 hour endurance run schedule (cont'd)

Run	Time (hours)	Engine Speed (rpm)	Load (Observed) hp	Coolant Temp. °F
130	1	650	0	180
131	4	3500	WOT	180
132	1	650	0	180
133	4	3500	52	180
134	1	650	0	180
135	4	3500	WOT	180
136	1	650	0	180
137	4	3500	52	180
138 *	1	650	0	180
139	4	4000	WOT	180
140	1	650	0	180
141	4	4000	53	180
142	1	650	0	180
143	4	4000	WOT	180
144	1	650	0	180
145	4	4000	53	180
146	1	650	0	180
147	4	4000	WOT	180
148 *	1	650	0	180
149	4	4000	53	180
150	1	650	0	180
151	4	4000	WOT	180
152	1	650	0	180
153	4	4000	53	180
154	1	650	0	180
155	4	4000	WOT	180
156	1	650	0	180
157	4	4000	53	180
158 *	1	650	0	180
159	Power check run - see paragraph 3.5.1			
160	1	650	0	200
161	2	4200	53	200
162	2	4200	WOT	200
163	1	650	0	200
164	2	4200	53	200
165	2	4200	WOT	200
166	1	650	0	200
167	2	4200	53	200
168	2	4200	WOT	200
169	1	650	0	200

* 25 Hour Intervals

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Table I. 500 hour endurance run schedule (cont'd)

Run	Time (hours)	Engine Speed (rpm)	Load (Observed) hp	Coolant Temp. °F
170	2	4200	53	200
171	2	4200	WOT	200
172	1	650	0	200
173	2	4200	53	200
174 *	2	4200	WOT	200
175	1	650	0	200
176	2	4200	53	200
177	2	4200	WOT	200
178	1	650	0	200
179	2	4200	53	200
180	2	4200	WOT	200
181	1	650	0	200
182	2	4200	53	200
183	2	4200	WOT	200
184	2.5	1800	WOT	200
185	2.5	1800	32	200
186	2.5	1800	WOT	200
187	2.5	1800	32	200
188	2.5	1800	WOT	200
189	2.5	4000	53	200
190	2.5	4000	WOT	200
191	2.5	4000	53	200
192	2.5	4000	WOT	200
193	2.5	4000	53	200
194	0.75	4200	WOT	200
195	0.25	4200	Motoring	200
196	0.75	4200	WOT	200
197	0.25	4200	Motoring	200
198	0.75	4200	WOT	200
199	0.25	4200	Motoring	200
200	0.75	4200	WOT	200
201	0.25	4200	Motoring	200
202	0.75	4200	WOT	200
203	0.25	4200	Motoring	200
204	0.75	4200	WOT	200
205	0.25	4200	Motoring	200
206	0.75	4200	WOT	200
207	0.25	4200	Motoring	200
208	0.75	4200	WOT	200
209	0.25	4200	Motoring	200

* 25 Hour Intervals

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Table I. 500 hour endurance run schedule (cont'd)

Run	Time (hours)	Engine Speed (rpm)	Load (Observed) hp	Coolant Temp. °F
210	0.75	4200	WOT	200
211	0.25	4200	Motoring	200
212	0.75	4200	WOT	200
213 *	0.25	4200	Motoring	200
214	0.75	4200	WOT	200
215	0.25	4200	Motoring	200
216	0.75	4200	WOT	200
217	0.25	4200	Motoring	200
218	0.75	4200	WOT	200
219	0.25	4200	Motoring	200
220	0.75	4200	WOT	200
221	0.25	4200	Motoring	200
222	0.75	4200	WOT	200
223	0.25	4200	Motoring	200
224	0.75	4200	WOT	200
225	0.25	4200	Motoring	200
226	0.75	4200	WOT	200
227	0.25	4200	Motoring	200
228	0.75	4200	WOT	200
229	0.25	4200	Motoring	200
230	0.75	4200	WOT	200
231	0.25	4200	Motoring	200
232	0.75	4200	WOT	200
233	0.25	4200	Motoring	200
234	0.75	4200	WOT	200
235	0.25	4200	Motoring	200
236	0.75	4200	WOT	200
237	0.25	4200	Motoring	200
238	0.75	4200	WOT	200
239	0.25	4200	Motoring	200
240	0.75	4200	WOT	200
241	0.25	4200	Motoring	200
242	0.75	4200	WOT	200
243	0.25	4200	Motoring	200
244 *	10	4400	WOT	200
245 Power check run - see paragraph 3.5.1				

* 25 Hour Intervals

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3.3 Design and construction. Design and construction of engines furnished under this specification shall conform to Drawings 11613729 and 11660437.

3.3.1 Accessories and equipment. All electrical accessories and equipment, including wiring and electrical connections, shall conform to the applicable requirements of MIL-E-13856. Unless otherwise specified, (see 6.2) all electrical accessories and equipment shall be installed on the engine, properly adjusted and operated at 28 ± 0.5 V.D.C.

3.3.2 Interchangeability of parts. Component assemblies and parts of the engine shall be so designed and constructed that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced and adjusted without requiring modification.

3.3.3 Fuel. The engine shall perform as specified herein using fuel in accordance with the seasonal requirements of MIL-G-46015. The initial production engine 500-hour endurance test, however, shall be performed using fuel conforming to a special referee grade meeting the requirements of MIL-G-46015. Fuel conforming to VV-G-001690 may be used during the production break-in run.

3.3.4 Oil. The engine shall perform as specified herein using lubricating oil in accordance with seasonal requirements of MIL-L-2104 within the ambient temperature range of -10° to + 115°F, and MIL-L-10295 from -65° to 0 F. Oil conforming to type I, grade 30 of MIL-L-21260 may be substituted during engine build up and the production break-in and power check runs (see 3.5.1). Referee grade MB901 of MIL-L-2104 oil shall be used during the initial production 500-hour endurance test.

3.3.5 Coolant. The engine shall perform as specified herein using water and antifreeze compound conforming to O-A-548, arctic-type antifreeze compound conforming to MIL-A-11755, or water and corrosion inhibitor compound conforming to O-I-490.

3.3.6 Air cleaner. The engine shall perform as specified herein when using an air cleaner conforming to the applicable drawing and MIL-A-13488.

3.4 Production break-in run. All engines offered for acceptance under this specification shall be subjected to the production break-in run as follows:

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Table II. Production break-in run

Period	Duration (minutes)	Engine Speed (rpm)	Intake Manifold Vacuum (in. hg)
1	as required	650 (Initial Adjustments)	N/A
2	10	2,000	16
3	10	3,000	10
4	as required	650	N/A
5	10	3,500	7
6	10	4,000	5
7	5	650	N/A

During each period of the break-in run the following data shall be recorded:

(a) Oil sump temperature (at drain plug).
(b) Gallery oil pressure (PSI).
(c) Engine speed (RPM).
(d) Period duration.
(e) Intake manifold vacuum (in. hg).
(f) Coolant temperature at inlet and outlet (the minimum cooling temperature shall be 140°F during break-in run).

All adjustments shall be made during break-in run. The valve lash shall be readjusted on a hot engine during run #4 immediately following run #3.

3.5 Performance requirements.

3.5.1 Power check run. Each engine submitted under this specification shall meet the performance requirements of this specification when tested in accordance with table III.

Table III. Power check run

Period	Engine Speed	Throttle Position
1	4,000	WOT
2	3,600	WOT
3	3,200	WOT
4	2,800	WOT
5	2,400	WOT
6	1,800	WOT
7	1,400	WOT
8	1,000	WOT

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3.5.2 Brake horsepower. The engine shall develop not less than 65 brake horsepower at 4,000 revolutions per minute (rpm), corrected to standard atmospheric conditions, at the completion of the production break-in run (see 3.4). Brake horsepower to be obtained only during initial production tests and control tests.

3.5.3 Torque. Under full throttle, the engine shall develop at the flywheel, not less than 113 pound-feet of torque at 1,800 rpm, corrected to standard atmospheric conditions, at the completion of the production break-in run (see 3.4). Torque to be obtained only during initial production tests and control tests.

3.5.4 Speed range. The engine shall operate satisfactorily under all loads and conditions as specified herein through a speed range of 650 to 4,000 rpm and shall maintain satisfactory idle speed of 650 rpm. At idle, the minimum manifold vacuum shall be 17 inches of mercury with a maximum fluctuations of 0.2 in. hg as measured on a manometer.

3.5.5 Fuel consumption. The fuel consumption of the bare engine shall not exceed the following limits when operating at maximum power output:

Maximum Observed bhp RPM	Specific Fuel Consumption lbs/bhp-hr
4,000	0.62
1,800	0.63

3.5.6 Oil consumption. After 25 hours of operation, the engine shall consume not more than 0.004 lbs. of lubricating oil per observed brake-horsepower hour, when operating at full load throughout the speed range of the engine with oil sump temperatures of 140° to 250°F, using oil as specified in 3.3.4.

3.5.7 Oil pressure. Under all operating conditions specified herein, including one pint low oil sump conditions, with an oil temperature of plus 140° to 250°F, gallery oil pressure shall be not more than 45 pounds per square inch (psi) nor less than 35 psi measured at the pressure sending unit, when the engine is operating at 3,000 rpm using oil as specified in 3.3.4 and shall not be less than 15 pounds per square inch (psi) when engine is idling at 650 rpm.

3.5.8 Compression pressure variation. Pressure variation between cylinders at cranking speed on a hot engine shall not exceed 15 psi at the conclusion of the 50-hour production control test (see 3.8) and 23 psi at the conclusion of the 500-hour endurance test (see table I).

3.5.9 Coolant temperature rise. The coolant temperature rise through the engine shall not exceed 12°F from inlet to outlet when the engine with coolant thermostat fixed in full open position is operating at wide open throttle and 4,000 rpm.

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3.6 Environmental requirements.

3.6.1 Environmental performance. The engine shall start and demonstrate the performance characteristics specified herein under all of the following conditions and possible combination of conditions:

(a) Without external aids after being thoroughly cold-soaked, without benefit of solar radiation, of ambient temperatures of minus 25°F. (Cold-soak shall be defined as reducing the temperature of the engine and its fuel and lubricating oil, to within 5°F of the ambient air temperature surrounding the engine).
(b) Without external aids, when operated at the follwing temperatures with exposure to maximum solar radiation:

Intake Cooling Air	140°F
Carburetor Intake Air	115°F
Ambient Air Temperature	115°F

3.6.2 Limiting operating temperatures. The engine shall not exceed the following limiting temperatures when operating at full throttle throughout the speed range of the engine under any of the conditions specified herein:

(1) Oil Sump 250°F (measured at the drain plug)
(2) Coolant 210°F (measured at coolant outlet).

3.6.3 Cold temperature (-25°F). The engine shall meet the performance requirements without external aids, shall be cold-soaked until its fuel, coolant and oil temperature are within ± 5°F of the ambient temperature. Without the aid of solar radiation, the engine shall be started and be subjected to the power check run as specified in 3.5.1 at -25°F ambient temperature.

3.6.4 Cold temperature (-65°F). The engine shall start at -65°F with the aid of Army Winterization Kit, #8712420.

3.6.5 High temperature (+115°F). The engine shall meet the performance requirements without external aids, shall be exposed to maximum solar radiation until its fuel, coolant and oil temperatures are within ±5°F of the ambient temperature. The engine shall be started and be subjected to the power check run as specified at 115°F ambient temperature, and as required in Para. 3.5.1.

3.6.6 Altitude. The engine shall be placed in an altitude chamber and the internal pressure reduced to 22.2 inches of mercury. The engine shall be started and subjected to the power check run as specified in Para. 3.5.1 to determine that the engine shall deliver not less than 70% of the minimum horsepower at sea level.

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3.6.7 Humidity conditions. When tested, the engine shall meet the performance requirements specified under relative humidity conditions as low as 5 percent at a temperature of 115°F and as high as 100 percent at all temperatures from plus 25° to plus 85°F. The engine shall be started and be subjected to the power check run as specified in Para. 3.5.1.

3.6.8 Grades and slopes. The engine shall demonstrate the performance characteristics specified when operated throughout its power and speed range. The engine shall operate at full and one pint low oil level conditions in the sump, on longitudinal grades up to 60 percent and on lateral slopes up to 40 percent and then be subjected to a power check run as specified in Para. 3.5.1.

3.6.9 Electromagnetic interference. The engine shall meet the requirements of MIL-STD-461 for Class 11C Equipment, and Test Procedures CE03, and RE02 of MIL-STD-462.

3.6.10 Submersion. The engine, with intake and exhaust ducted to the atmosphere, no fan blade attached and with applicable crankcase fording valve installed, shall be operated at 2000 ± 500 rpm for a period of 30 minutes while submerged to a depth of 24 inches above the valve housing in either fresh or salt water, and while still submerged shall restart after being stopped for three minutes and then operated for an additional fifteen minutes.

3.7 Steam and water jet cleaning. The engine and all its components shall withstand cleaning and high pressure steam and cleaner conforming to MIL-C-52412 and water jet, without deterioration of seals or hose, water leakage past seals or gaskets, or other defects. Paint removal shall not be condidered a deficiency. Steam and water jet cleaning is defined as follows:

> The jet is applied perpendicular to the surface being cleaned at a distance of not more than one foot from the surface for steam cleaning, and not more than five feet from the surface for water jet cleaning, and cleaned at a rate of one square foot per minute. The jet pressure shall be not less than 100 psi and not more than 110 psi.

3.8 50-hour production control test. Engines selected for control test shall meet the performance requirements of this specification when tested in accordance with tests specified in Para. 4.5.1, table VII. For the purpose of assuring the continuing quality of engines being produced under the contract, a production control test of 50 hours duration shall be conducted by the contractor (see table IV). All data required during this test shall be obtained under stabilized operating conditions. The engine shall be coupled to a suitable dynamometer. At each start, the engine shall be warmed-up until torque, speed and temperature have been stabilized for a period of not less than one minute. All applicable temperatures shall be recorded in the test log at the conclusion of each warm-up period. Warm-up, starting, and stopping times shall be recorded however,

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warm-up time and running time of less than 30 minutes shall not be counted toward fulfillment of test run hour requirements. The 50-hour duration of this test shall be run as near continuously as possible. NOTE: If oil weighing equipment is not used, the oil level in the engine sump shall be checked at 4-hour intervals. Necessary additions to maintain the oil level at the "full" mark on the dip stick, shall be made in one pint increments and the additions recorded in the test log sheet.

Table IV. 50-hour production control test

Period	Time (Hours)	Speed (rpm)	Load (hp)
1	(Production Break-In Run - See Table II)		
2	2	2,000	6
3	2	3,000	11
4	2	3,600	11
5	2	3,600	WOT
6	4	1,500	WOT
7	4	2,300	WOT
8	4	3,000	WOT
9	4	3,200	WOT
10	26	4,000	WOT
11	Power Check Run - See Para. 4.5.2		

3.8.1 Required data. During performance of the 50-hour production control test the following data shall be recorded in the test log sheet at one hour intervals:

(a) Engine speed (rpm).
(b) Observed brake horsepower (obs bhp).
(c) Observed torque (lbs ft).
(d) Inlet manifold vacuum (in. Hg).
(e) Fuel pressure (carburetor - psi).
(f) Lubricating oil pressure (gallery - psi).
(g) Lubricating temperature (engine oil sump - ^{o}F).
(h) Coolant temperature (engine inlet and outlet - ^{o}F).
(i) Coolant temperature rise (engine inlet to outlet - ^{o}F).
(j) Air temperature (carburetor inlet - ^{o}F).
(k) Air temperature (test cell ambient - ^{o}F).
(l) Exhaust temperature (exhaust manifold outlet - ^{o}F).
(m) Fuel rate (lbs/hr).
(n) Brake specific fuel consumption (lbs/observed bhp hr).
(o) Lubricating oil consumption (lbs/hr).
(p) Brake specific oil consumption (lbs/observed bhp hr).

3.8.2 Atmospheric conditions. During performance of the 50-hour production control test, observed barometric and psychrometric readings shall be recorded at 4-hour intervals and immediately preceding the power run.

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3.8.3 Performance. Engines subjected to the 50-hour production control test shall, after completion of the test, conform in all respects to the performance requirements specified herein for all production engines.

3.8.4 Failure and corrections. Upon completion of the 50-hour production control test, the engine shall be completely disassembled and all parts inspected. A complete report including a copy of the test log sheet and the results of the post-test inspection shall be submitted to the Government quality assurance representative and a copy to the contracting officer. Should there be any evidence of malfunction or parts failure as a result of this test, the Government quality assurance representative shall refuse acceptance of subsequent production engines until objective evidence, to his satisfaction, has been presented by the contractor that corrective action has been instituted in production. After inspection, the engine shall be reassembled using original parts that are in good condition and new parts to replace those found unsatisfactory during examination.

3.8.5 Disposition of 50-hour production control test engines. The engine, after reassembly and correction of all defects disclosed by the test, shall be given a break-in run, inspection and adjustment, and power check run following the procedures specified herein. If found to be acceptable, shipped as a regular production engine. The break-in run can be omitted if all of the original parts are reused.

3.9 Log sheet. The contractor shall maintain a log sheet of each engine run during all tests specified herein. The form of the log sheet and number of legible copies to be furnished to the Government shall be as approved by the procuring activity.

3.10 Exterior surface treatment. All exposed exterior surfaces of the engine and its components, with the exception of the exhaust manifolds, shall be cleaned, painted or treated for corrosion resistance as specified on the applicable drawing or, if not so specified, in accordance with the applicable provisions of MIL-STD-193.

3.11 Identification of product.

3.11.1 Marking. The engine shall be marked in accordance with MIL-STD-130, unless otherwise shown on applicable drawings.

3.11.2 Nameplate. The engine shall have a nameplate attached to the engine in a conspicuous location, so that plate may be easily read when the engine is installed in the vehicle. The nameplate shall conform to MIL-P-514 and shall contain the following minimum information:

Engine - 65 HP, 4-cylinder-in-line
Military part number
Federal stock number
Manufacturer's identification
Manufacturer's serial number
US

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractors quality control system. Unless otherwise specified by the procuring activity (see 6.2), the contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure or other applicable documents referenced herein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to the Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to determine quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product, will immediatley be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Parts, components and subassemblies. Parts, components and subassemblies shall be inspected in accordance with drawings and applicable specifications (refer to para. 3.3 thru 3.3.6 and para. 3.10) and with the supplementary quality assurance provisions provided in Index SQAP 11613729.

4.2 Initial production engine inspection. The initial production engine produced shall be subjected to the examination for defects specified in table VI and the tests specified in table V. Initial production inspection shall be conducted by the Government at a place designated by the Government. To determine conformance to 3.1 and 3.2 thru 3.2.10 the initial engine of any contract lot, shall be subjected to a laboratory endurance test of 500-hours duration, according to the schedule in table I.

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Table V. Order of initial production testing

Paragraph No.	Test
3.2	500-hour endurance test
3.6	Environmental

4.2.1 Failure. Failure of test engines to comply with requirements specified or defects of a workmanship or material nature, revealed during or as a result of test, shall be cause for rejection of engines. Further, the Government may refuse to continue acceptance of production engines until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defect found during or as a result of this test shall be prima facie evidence that all engines already accepted prior to completion of test, are similarily defective unless evidence satisfactory to the Government is furnished by the contractor that they are not similarily defective. Such defects on all engines shall be corrected by the contractor at no cost to the Government.

4.3 Production acceptance requirements.

4.3.1 Sampling for engine examinations.

4.3.1.1 Lot formation. A lot shall consist of all engines of one type from one day's production submitted at one time for acceptance inspection.

4.3.1.2 Sample size. Sample size is the number of units drawn from a lot to be submitted fo quality conformance examination. The sample size shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each engine selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective for majors and defects per hundred units for minors when inspecting samples from each lot of engines.

Classification	AQL
Major	6.5
Minor	25

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified by the characteristic as shown in table VI

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Table VI. Classification of defects

Categories	Characteristics	Defects	Gage or other method of inspection
Major:			
101	Compression pressure	Excessive variation between cylinders	Gage
102	Valve tappet clearance each valve, engine hot	Intake and exhaust valves improperly adjusted	Gage
103	Oil sump, and oil filters	Dirt, chips, and foreign particles, due to production	Visual
104	Fuel, oil, and coolant leaks	Leakage	Visual
105	Fuel lines, shut-off valves	Damaged	Visual & functional
106	Torque on cylinder head bolts, intake and exhaust manifold flange bolts	Improper torque	Torque wrench
107	Crankshaft	Excessive or restrictive end play	Gage & functional
108	Engine	Malfunction	Functional
109	Fuel system components	Malfunction - damage or leaks, etc.	Visual & functional
110	Ignition system, timing & components	Malfunction, improper adjustment, damaged or defective wiring, etc.	Visual & functional
Minor:			
201	Linkage	Improperly adjusted	Visual
202	Minor assemblies	Omitted	Visual

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Table VI. Classification of defects (cont'd)

Categories	Characteristics	Defects	Gage or other method of inspection
Minor:			
203	Engine	Improper adjustment or installation of components	Visual & functional
204	Fuel lines, hose vents	Improper assembly or installation	Visual
205	Painting	Spots missed, sags, or runs	Visual
206	Workmanship	Not following good practice, improper installation of components, improper adjustment of components	Visual & functional

Acceptability of a lot or batch will be determined by the use of a sampling plan or plans with the designated AQL or AQL's.

4.3.3 Acceptance testing.

4.3.3.1 Testing. Each engine shall be subject to the production break-in run as specified in paragraph 3.4.

4.3.3.2 Failure. Failure of an acceptance test sample to pass any specified examination or test may be cause, at the option of the Government, to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.4 Control tests.

4.4.1 Sampling for production control tests. Control test samples shall be selected at the rate of one engine for every 100 produced (or portion thereof), except that not more than three engines shall be selected in any 30 day period. Samples shall be from a production lot which has passed the quality conformance examination specified in 4.3.2 and the break-in run as specified in table II. The 50 hour control test shall be conducted as specified in paragraph 3.8 thru 3.8.5 and tests listed in table VII.

95

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4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause, at the option of the Government, to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.5 Test procedures.

4.5.1 General. Tests are to be accomplished in accordance with table VII.

Table VII. Test schedule

Paragraph Reference	Test	Initial Production Test	Government Examination Test	Government Control Test
3.4	Break-in	X	X	X
3.5.1	Power Check	X	-	X
3.8	50 Hour Control	-	-	X
3.2	500 Hour Endurance	X	-	-
3.6	Environmental	X	-	-
4.3.2	Examination	X	X	X

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. The ambient room temperature of the laboratory shall be 77° plus or minus 15°F. Unless otherwise specified herein or in applicable specifications, specimens shall be tested at the ambient room temperature of the laboratory. Engines shall be heat stabilized for a period of one hour prior to being subjected to any test excepting that the one hour stabilization period shall be omitted for consecutive tests at the same ambient temperature. Tests results shall be corrected to standard sea level conditions.

4.5.1.2 Apparatus. The apparatus shall consist of a dynamometer and its basic components.

4.5.1.3 Calibration of test equipment. Unless otherwise specified herein, test equipment accuracy shall be as to permit measurement of 10 percent of product on test specification tolerance. Calibration of test equipment shall be conducted at intervals to establish required accuracy. Records of calibration shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Required measuring instruments tolerances are as listed in table VIII.

95

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Table VIII. Measuring instrument tolerances

Instrument	Accuracy of Tolerance
Dynamometer	5 percent
Voltimeter	2 percent
Temperature Gage	2 percent

4.5.2 Power check run. Full throttle performance characteristics of the engine shall be determined by operating the engine according to Para. 3.5 thru 3.5.9. All data necessary to determine corrected horsepower and torque and specific fuel consumption shall be recorded in the log. Performance curves showing corrected horsepower and torque and specific fuel and oil consumption shall be required in addition to the log sheets. Oil consumption data shall be obtained from endurance hours. All data shall be taken after stabilized operating conditions have been obtained.

4.5.3 Environmental.

4.5.3.1 Cold temperature test (-25°F). The engine shall be started and be subjected to the power check run as specified in 3.5.1 at -25°F ambient temperature to determine conformance to Para. 3.6.3 requirements.

4.5.3.1.1 Cold temperature test (-65°F). The engine shall be started and be subjected to the power check run as specified in 3.5.1 at -65°F ambient temperature to determine conformance to Para. 3.6.4 requirements.

4.5.3.2 High temperature test (plus 115°F). The engine shall be started and be subjected to the power check run as specified in 3.5.1 at 115°F ambient temperature to determine conformance to Para. 3.6.5 requirements.

4.5.3.3 Altitude test. The engine shall be started and be subjected to the power check run as specified in 3.5.1 to determine conformance to Para. 3.6.6 requirements.

4.5.3.4 Humidity. The engine shall be started and be subjected to the power check run as specified in 3.5.1 to determine conformance to Para. 3.6.7 requirements.

4.5.3.5 Grades and slopes. The engine shall be started and demonstrate the performance characteristics specified in Para. 3.6.8 and then be subjected to a power check run as specified in Para. 3.5.1 requirements.

4.5.3.6 Electromagnetic interference. To determine conformance to Para. 3.6.9, the engine shall be tested in accordance with the applicable provisions of MIL-STD-461, Equipment Class 11C and Test Procedures CE03 and RE02 of MIL-STD-462.

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4.5.3.7 Submersion. The engine shall meet the requirements specified in Para. 3.6.10.

4.6 Log sheets. Log sheets shall be complete and made available to the Government in accordance with Para. 3.9 requirements.

4.7 Inspection of preparation for delivery.

4.7.1 Materials and processes. All materials and processes involved in the preparation for delivery shall be inspected to determine conformance to requirements of section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking for the desired level of protection shall be in accordance with applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The engine covered by this specification is intended for production use, or as a replacement in military tactical and administrative vehicles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Parts and accessories other than as specified (see 3.3.1).
(c) Number of log sheets required (see 3.9).
(d) If contractor is to maintain a quality control system (see 4.1.1).
(e) Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Initial production engine. The initial production engine requirements may be waived for those manufacturers who supplied engines conforming to this specification during the past two years.

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6.4 Correction of observed data. Values of torque and horsepower of engines should be corrected for standard atmospheric conditions as follows:

$$\text{Corrected BHP} = \text{observed BHP} \times \frac{29.92}{B-E} \sqrt{\frac{460 + T}{520}}$$

$$\text{Corrected torque} = \frac{\text{corrected BHP} \times 5252}{\text{RPM}}$$

B = true barometer, In. Hg.
E = water vapor pressure (from humidity), In. Hg.
T = intake air temperature, °F at the inlet to the air cleaner
BHP = brake horsepower

6.5 Motoring. Motoring is defined as driving the engine with the dynamometer at the speed specified, with the throttle at idle position, the fuel line open, and the ignition on. This condition simulates engine braking characteristics during vehicle deceleration or down hill operation.

6.6 Brake horsepower. Brake horsepower shall be defined as the horsepower output of the engine at the flywheel.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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